

Figure 7 / Section 5

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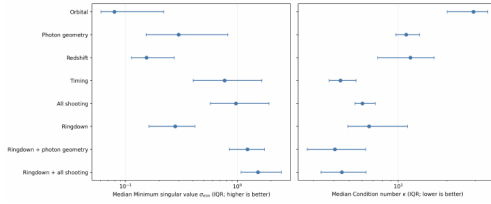


Figure 7: Grid-wide median and interquartile summaries of $\sigma_{\min}$ and condition number. Photon geometry provides the clearest complement to ringdown.

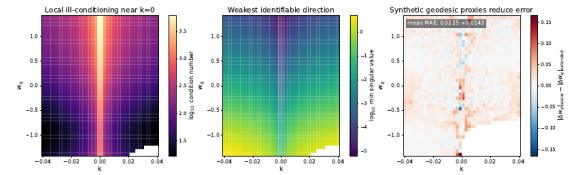


Figure 1: Historical analytic identifiability lens motivating the physical-shooting study. Left: local Jacobian condition number, which grows near $k = 0$. Center: minimum singular value, showing the weakest identifiable direction. Right: change in identifiable- w_q error after adding the historical synthetic geodesic proxies; the mean MAE changes from 0.0225 to 0.0143. White regions are invalid or missing source-grid points and are not interpolated. The first two panels diagnose inverse-map geometry; the third records the original proxy experiment and must not be interpreted as physical ray tracing or geodesic shooting. Proxy augmentation does not resolve the exact $k = 0$ rank loss.

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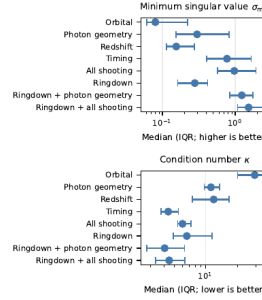


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Normalized mean absolute error divides MAE by the full target span. Split-conformal intervals target 90% coverage and are interpreted under their exchangeability assumptions [22, 20, 1]. Feature-noise scales and ev-

5 Leakage-Aware Learning Protocols

Five primary feature sets are ordered consistently: ringdown, photon geometry, all shooting, ringdown plus photon geometry, and ringdown plus all shooting. Here photon geometry and the other integrated quantities are physical shooting observables, distinct from the historical synthetic geodesic proxies. Models are histogram gradient boosting (HGB), random forest (RF), and a target-scaled multilayer perceptron (MLP) [4, 9, 17]. The protocols are random interpolation, two blocked 3×3 grouped physical layouts, and four directional extrapolation tests. All nominal $k = 0$ records are retained and flagged, but ordinary w_q error excludes them because w_q is exactly non-identifiable there.

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6 Inverse-Learning Results

Random splitting is optimistic. For HGB, RF, and MLP considered together, random ringdown NMAE is 0.0361/0.1688 for k /identifiable w_q , while grouped ringdown is 0.0485/0.3093. Ringdown plus photon geometry gives grouped 0.0457/0.0875, with the w_q improvement reproduced independently in both tree families. Directional extrapolation remains harder: the combined set has aggregate NMAE 0.1305/0.2012. Figure 8 compares

Inverse-

the three validation regimes directly.

Conditioning and coverage explain different aspects of error. In the error model, standardized training distance has estimate 0.330, extrapolation status 0.276, and log condition number 0.197. Physical conditioning generally agrees with grouped tree-based w_q performance, but it does not perfectly rank every model, target, or protocol. The MLP supplies a counterexample: it can have reasonable central metrics while producing catastrophic directional tails. The aligned physical and model diagnostics appear in Fig. 9.

Nominal 90% conformal coverage is 0.892/0.965 for random k/w_q , but falls to 0.722/0.851 under grouped interpolation and 0.509/0.660 under extrapolation. This is empirical coverage under shift, not a retained finite-sample guarantee. Figure 10 separates coverage by target and protocol.

7 Numerical Convergence and Estimator Robustness

The uniform 161-phase study completes all 121 physical points. From 9317 feature-point comparisons, 240 are unresolved under the frozen criterion; the median normalized feature change is 7.908×10^{-4} and the maximum is 0.08069. Across frozen scored predictions, the median normalized shift is 9.482×10^{-4} and the 95th percentile is 0.02606. The median aggregate NMAE change is 6.880×10^{-4} , but the maximum is 16.8497 because of un-clipped MLP directional extrapolation outliers.

The targeted 321-phase audit selects 35 points deterministically before reading their 321-phase outcomes. All complete. Median and 95th-percentile normalized 161-to-321 feature changes are 1.963×10^{-4} and 7.234×10^{-3} ; no comparison remains unresolved, ringdown is exactly unchanged, and Jacobian rank is preserved. Forward convergence therefore passes. Figure 11 displays both resolution increments on the same selected-point order.

Frozen trees remain tail-sensitive: HGB and RF 95th-percentile normalized shifts are 0.08122 and 0.08509, and their largest identifiable- w_q shifts are 0.30593 and 0.27053. Their median shifts are zero because many tree predictions do not cross a split threshold. The predeclared perturbation ensembles and fixed Extra Trees smoother do not pass all stability and grouped-performance criteria. Of 410 catastrophic MLP records, 409 are already catastrophic at 161 phases, none is materially worsened by the 321-phase substitution, 63 reach the iteration limit, and no numerical overflow is detected. The decision is Outcome B: the forward map converges, but estimator robustness does not.

Typical predictions are stable, but the tails are not. The earlier MLP maxima are not representative of every estimator, while the targeted tree failures show that the remaining issue also cannot be attributed solely to the MLP. Journal readiness remains conditional.

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Figure 12 separates the estimator mechanisms. Tree medians remain exactly unchanged because most perturbations do not cross a split, yet their tails do. The MLP reacts more smoothly to phase substitution, while its catastrophic directional records overwhelmingly predate that substitution.

8 Discussion

Jacobian conditioning answers whether nearby parameter directions are separated by the chosen observables. Training distance answers whether a fitted estimator has support where it is evaluated. Neither statistic subsumes the other. Consequently, $\sigma_{\min}$ can predict broad observable-set ordering without ranking every empirical model or protocol.

Tree ensembles are piecewise and can show zero median response followed by large split-crossing tails. The

Figures 11-12 / conclusion

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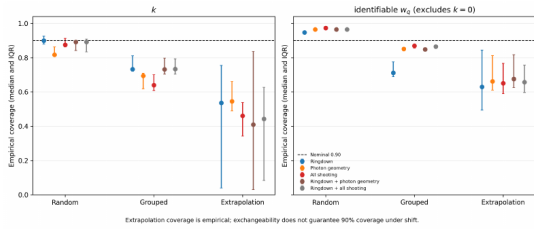


Figure 10: Empirical 90% split-conformal coverage by target and protocol. Ordinary w_q coverage excludes $k = 0$; extrapolation coverage is not guaranteed under distribution shift.

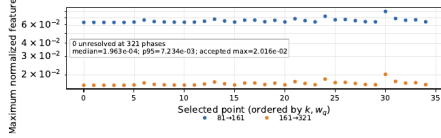


Figure 11: Targeted pointwise maximum normalized feature changes from 81 to 161 and 161 to 321 phases. The 321-phase audit resolves the forward-feature criterion but does not by itself certify inverse estimators.

synthetic. Although the targeted audit resolves higher-harmonic forward differences on its selected set, it is not a uniform 321-phase grid and tree tail sensitivity persists. MLP extrapolation is unstable. The verdict is therefore conditional, and no result is an observational constraint.

10 Conclusion

Physical complementarity is supported: ringdown constrains k , while photon geometry supplies an independent direction needed for much of the w_q recovery. Random interpolation overstates recoverability relative to grouped and directional validation. Finally, numerical and distribution-shift audits are necessary before confident inverse claims: the forward features can converge while fitted estimators remain fragile. These conclusions motivate robust inverse methods and broader physical modeling before observational use.

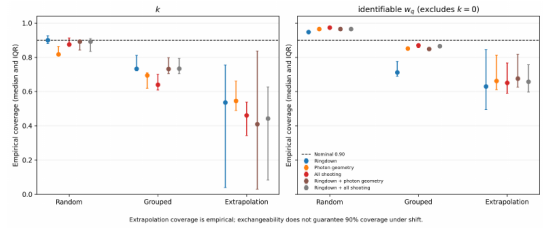


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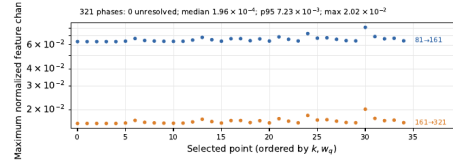


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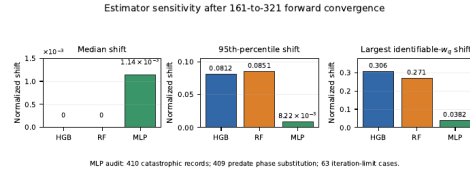


Figure 12: Frozen-estimator sensitivity after forward convergence. Bars show the median, 95th-percentile, and largest identifiable- w_q normalized prediction shift for HGB, RF, and MLP. The audit note summarizes the distinct MLP failure audit. No predictions are clipped.

Arrival appendix

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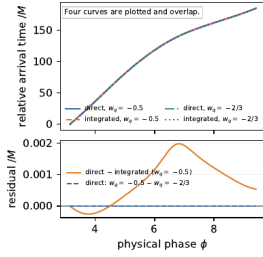


Figure 14: Schwarzschild arrival-time validation. Four arrival-time curves are plotted. The two nominal w_q cases coincide because the $k = 0$ geometry is Schwarzschild, while the direct and integrated-redshift constructions agree within numerical quadrature error. The lower panel shows direct minus integrated arrival time for $w_q = -0.5$ and the direct-arrival difference between $w_q = -0.5$ and $w_q = -2/3$ on their actual numerical scales.

repair distribution shift.

F Noise and Learning Curves

Grouped one-percent feature noise yields ringdown-plus-photon-geometry NMAE 0.0699/0.0853 for k/w_q . Learning curves are still improving at 78 training points, so data limitation remains part of the interpretation.

G Numerical-Validation Thresholds

The 35-point 321-phase run takes 5318.2 s. Maximum hit, timelike-constraint, null-constraint, and impact-parameter drift errors are respectively 7.0913×10^{-8} , 3.1303×10^{-13} , 7.6214×10^{-12} , and 1.7932×10^{-11} . The Schwarzschild arrival-time comparison in Fig. 14 separates the visually overlapping curves from their actual-scale residuals.

Matched Schwarzschild and Kiselev direct rays are compared on their actual coordinate scale in Fig. 15.

Table 2: Targeted 321-phase frozen-estimator robustness.

Model	median	95th percentile	max. w_q
HGB	0	0.08122	0.30593
RF	0	0.08509	0.27053
MLP	0.001143	0.008223	0.03819

H Uniform 161-Phase Convergence Audit

All 121 points complete in 4473.2 s with zero failed points. The audit preserves the exact $k = 0$ boundary and uses no mixed-resolution row, interpolation, or proxy substitution.

I MLP Extrapolation-Failure Audit

The machine-readable table contains standardized input magnitude, hidden activation magnitude, iteration status, overflow flag, training distance, and resolution response for every catastrophic record. It distinguishes optimization failure from extrapolation instability without clipping. Exact frozen-estimator shift summaries used in Fig. 12 are retained in Table 2 for numerical reference.

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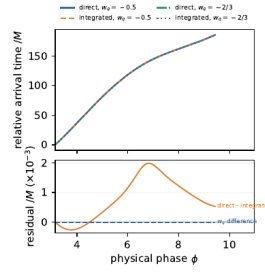


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